Predicting Individual Reaction Time from Task-fMRI Functional Connectivity

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***Abstract*—Predicting behavioral performance from brain activity remains a major challenge in cognitive neuroscience due to the high dimensionality and complexity of neuroimaging data. This study investigates whether task-evoked functional MRI (tfMRI) connectivity patterns can predict individual reaction time (RT) during the Human Connectome Project (HCP) Working Memory task. Functional connectivity features were derived by parcellating cortical BOLD signals into 360 regions of interest using the Glasser atlas, collapsing these into 12 large-scale functional networks, and computing pairwise Pearson correlations to yield a compact 78-feature connectivity vector per subject. Elastic Net regression was selected as the primary predictive model for its ability to simultaneously perform feature selection and coefficient shrinkage through combined L_1 and L_2 regularization, addressing the multicollinearity inherent in network-level fMRI data. Recursive Feature Elimination (RFE) was further applied to identify the most predictive connectivity edges, and nested cross-validation was used to ensure robust out-of-sample generalization. The model achieved a test-set correlation of $r \approx 0.21$, consistent with the range reported in the connectome-based predictive modeling literature for working memory tasks. Beyond prediction accuracy, this work emphasizes model explainability by mapping learned beta weights onto a network adjacency heatmap, a 2D connectome, and a 3D cortical surface projection. This multi-level interpretability analysis identifies the Prefrontal Cortex and Posterior Cingulate Cortex as primary neural bottlenecks of cognitive processing speed, and highlights the balance between visuolinguistic facilitation and somatomotor interference as the key determinant of individual differences in 2-back reaction time.**

***Index Terms*—fMRI, HCP, working memory, reaction time, machine learning, Elastic Net, Ridge Regression**

I. Introduction

FUNCTIONAL Magnetic Resonance Imaging (fMRI) is used to study brain activity during cognitive tasks by measuring changes associated with blood oxygenation levels. Task-based fMRI experiments allow researchers to examine how different brain regions respond while participants perform tasks involving memory, attention, language, and decision-making. One of the largest publicly available datasets developed for this purpose is the Human Connectome Project (HCP), which provides standardized neuroimaging and behavioral data collected across multiple cognitive experiments.

This work focuses on the HCP Working Memory task, which combines working memory processing with category-specific visual stimuli. During the experiment, participants viewed images belonging to four categories: faces, places, tools, and body parts. The task included two experimental conditions known as the 0-back and 2-back tasks.

In the 0-back condition, participants were shown a target image at the beginning of a block and were instructed to respond whenever the same image appeared during the following trials. This condition mainly involved target recognition and sustained attention. In the 2-back condition, participants were required to determine whether the current image matched the image presented two trials earlier. Since the target changed continuously throughout the block, participants had to continuously update and compare information in working memory, making the task cognitively more demanding than the 0-back condition.

The experiment followed a block-based design. Each run contained separate blocks for the four stimulus categories and included both 0-back and 2-back conditions. At the beginning of each block, a 2.5-second cue indicated the task type and, in the 0-back condition, displayed the target stimulus. Each task block consisted of 10 trials, where every trial lasted 2.5 seconds. During a trial, the stimulus was displayed for 2 seconds followed by a 0.5-second inter-task interval. Fixation blocks were also included between task blocks to provide baseline measurements of brain activity.

While participants performed the experiment inside the MRI scanner, both neural activity and behavioral responses were recorded. The released HCP data includes task timing files, event-related variables, reaction time (RT), and response accuracy measurements for the working memory task. In addition, the dataset provides explanatory variable (EV) files containing onset timings for different task conditions, including correct trials, incorrect trials, and no-response trials for both the 0-back and 2-back conditions.

Task-based functional connectivity features, representing the temporal synchronization between 12 large-scale functional networks, were used in this study as input features for machine learning models. By collapsing the 360-region Glasser cortical atlas into functional networks, we reduced the dimensionality of the neural data while capturing stable, system-wide interactions. Since neuroimaging data is inherently high-dimensional and exhibits significant multicollinearity, traditional regression methods often suffer from instability and overfitting. To address these challenges,

this work applies a regularized Elastic Net regression within a Recursive Feature Elimination (RFE) wrapper. This hierarchical approach is designed to handle high-dimensional datasets and perform automated feature selection, ensuring the model remains robust while maintaining a high degree of interpretability.

The main objective of this study is to investigate whether these network-level connectivity fingerprints can predict individual reaction time (RT) during the HCP 2-back working memory task. Beyond simple prediction performance, this study implements a "Model Explainability" framework, as advocated by [8], to identify the specific neural bottlenecks—the functional connections that contribute most strongly to reaction time variability across participants. Through this approach, the work combines advanced predictive modeling with a physiological analysis of how coordination between executive and internal-facing brain networks determines cognitive processing speed.

II. Literature Review

Recent studies have shown that functional connectivity patterns derived from fMRI can provide useful information about cognitive performance. Instead of analyzing isolated brain regions, many approaches examine interactions between distributed brain networks.

Finn *et al.* introduced the concept of *functional connectome fingerprinting*, demonstrating that individuals can be identified using whole-brain connectivity patterns **Finn2015**. Their work showed that connectivity profiles are stable across sessions and contain subject-specific information.

Rosenberg *et al.* later demonstrated that whole-brain functional connectivity could predict sustained attention performance across individuals **Rosenberg2016**. Their study supported the idea that distributed connectivity features can be used for behavioral prediction tasks.

Avery *et al.* applied connectome-based predictive modeling (CPM) to Human Connectome Project (HCP) working memory data **Avery2020**. Using functional connectivity features, they predicted 2-back working memory accuracy and found that interactions involving prefrontal and parietal networks were strongly related to performance. The study also noted that prediction scores in neuroimaging are often modest because behavioral traits are influenced by many distributed neural processes.

Several studies have discussed the challenges of predictive modeling in neuroimaging. Poldrack *et al.* highlighted issues such as high dimensionality, limited sample sizes, and weak effect sizes, which can lead to low or negative R^2 values in behavioral prediction tasks **Poldrack2020**. Similarly, Sui *et al.* reviewed machine learning methods for neuroimaging-based prediction and emphasized the difficulty of obtaining highly accurate behavioral predictions from functional connectivity alone **Sui2020**.

Shen *et al.* proposed a connectome-based predictive modeling framework that combines feature selection and cross-validation to reduce overfitting in connectivity-based prediction tasks **Shen2017**. In addition, He *et al.* compared deep learning approaches with traditional regression models and found that regularized linear models often perform similarly to more complex architectures for functional connectivity prediction **He2020**.

Motivated by these previous findings, the present study investigates whether network-level functional connectivity features extracted from HCP working memory task fMRI data can predict reaction time during the 2-back task condition.

III. Data Preprocessing

The study used the Neuromatch Academy (NMA) curated subset of the Human Connectome Project (HCP) task-fMRI dataset. The dataset contained working memory (WM) task recordings from 100 subjects. Each subject completed two runs of the experiment (LR and RL), and the fMRI data were already preprocessed and parcellated into 360 cortical regions of interest (ROIs) based on the Glasser atlas.

Several preprocessing steps had already been applied to the released HCP data before analysis. These included motion correction, spatial normalization, surface-based alignment, artifact removal, and projection into predefined cortical parcels. Instead of voxel-level signals, each subject's data were provided as ROI-level BOLD time series, reducing dimensionality while preserving large-scale brain activity patterns.

For each subject and run, the ROI time series were loaded from the `data.npy` files. The mean signal of each ROI was subtracted from its time series to remove baseline offsets and center the data around zero. Event timing information was obtained from the explanatory variable (EV) files provided with the dataset. These EV files contained the onset times and durations of each working memory condition.

To reduce feature dimensionality and improve interpretability, the 360 ROIs were grouped into functional brain networks using the provided network labels. For each network, the ROI signals belonging to that network were averaged to generate a single network-level time series. Pearson correlation coefficients were then computed between all pairs of network signals to create a functional connectivity matrix for each subject and run.

Only the upper triangular portion of the connectivity matrix, including the diagonal, was retained to avoid redundant features. This resulted in 78 connectivity features per run. The features from the two runs were averaged together to obtain a more stable subject-level representation. The final feature matrix is represented as:

$$X \in R^{100\times 78}$$

where each row corresponds to one subject and each column corresponds to a unique functional connectivity feature between brain networks.

Behavioral reaction time (RT) values were extracted from the `Stats.txt` files associated with the working memory task. The mean reaction time for the 2-back condition was calculated separately for each run and then averaged across both runs to obtain a single RT value per subject. These values formed the target vector:

$$y \in R^{100}$$

where each element represents the average 2-back reaction time for one subject.

The dataset was then divided into training and testing subsets using an 80/20 split. Feature standardization was applied using z-score normalization based only on the training data. The same scaling parameters were then applied to the test set to avoid data leakage.

IV. Methodology

1) Data Acquisition and Stimulus Paradigm : We utilized task-based fMRI (tfMRI) data from a subset of 100 healthy young adults from the Human Connectome Project (HCP) S1200 release. The behavioral focus was the N-back Working Memory (WM) task, which alternates between 0-back (baseline) and 2-back (high cognitive load) conditions across four stimulus categories: body parts, faces, places, and tools. The target variable (y) was defined as the individual's **Median Reaction Time (RT)** during 2-back trials. This metric was extracted from the behavioral performance records and averaged across both the left-to-right (LR) and right-to-left (RL) phase-encoding runs to ensure a stable representation of subject-specific processing speed.

2) Feature Engineering: From Regions to Networks : To address the high dimensionality and low signal-to-noise ratio (SNR) of voxel-wise data, we implemented a hierarchical feature extraction pipeline:

1) **Parcellation:** Cortical BOLD signals were aggregated into 360 Regions of Interest (ROIs) using the multimodal Glasser Atlas.
2) **Network Convergence:** To capture "trait-like" properties and reduce the risk of overfitting, these 360 ROIs were collapsed into 12 large-scale functional networks (e.g., Frontoparietal, Dorsal Attention, and Default Mode networks).
3) **Connectivity Mapping:** For each subject, we computed a 12 X 12 Pearson Correlation Matrix, representing the functional connectivity between these networks. The unique connections from the upper triangle were extracted, yielding a concise feature vector of p = 78 connectivity edges.

3) Predictive Modeling and Regularization : We employed Elastic Net Regression to model the relationship between neural connectivity (X) and reaction time (y). Elastic Net was chosen for its ability to perform simultaneous feature selection and coefficient shrinkage by combining L_1 (Lasso) and L_2 (Ridge) penalties:

$$\min_w \frac{1}{2n}\|y - Xw\|_2^2 + \alpha\rho\|w\|_1 + \frac{\alpha(1-\rho)}{2}\|w\|_2^2 \quad (1)$$

where α controls the overall penalty strength and ρ (the *l1_ratio*) determines the balance between sparsity and smoothness. This hybrid regularization approach is particularly effective for fMRI data, where high-dimensional brain networks often exhibit significant inter-correlation (multicollinearity). While the L_1 component promotes a sparse solution by zeroing out non-essential connections, the L_2 component ensures that groups of highly correlated networks are penalized collectively rather than being arbitrarily excluded.

4) Automated Feature Selection (Wrapper Method) : To further refine the model and identify the specific "bottlenecks" of cognitive processing, we implemented a **Recursive Feature Elimination (RFE)** wrapper. The RFE process followed an iterative prune-and-retrain cycle:

1) Train the Elastic Net model on the full connectivity set.
2) Rank features by their absolute coefficient magnitude.
3) Recursively remove the least predictive edges until a target subset of features is reached.

5) Hyperparameter Optimization and Validation : To ensure robust generalization, we utilized a **Nested Cross-Validation** framework. The inner loop performed a grid search to optimize the following parameters: α (Regularization strength): $[0.1, 1, 10]\rho$ (L1 Ratio): $[0.1, 0.5, 0.9]k$ (Number of RFE features): $[10, 20, 40]$Final model performance was assessed in the outer loop using the R^2 coefficient of determination and Mean Absolute Error (MAE), providing a conservative estimate of how well connectivity "fingerprints" predict behavior in unseen individuals.

V. Results and Discussion

A. Model Performance and Benchmarking

The Elastic Net regression model achieved a test-set coefficient of determination (R^2) of 0.0452, a Mean Absolute Error (MAE) of 92.56 ms, and a Root Mean Squared Error (RMSE) of 111.13 ms. Although a low absolute R^2 value might initially appear to indicate poor model performance, this result must be interpreted within the specific context of neuroimaging-based behavioral prediction. As highlighted by Poldrack *et al.* **Poldrack2020** and Sui *et al.* **Sui2020**, the prediction of continuous behavioral metrics from fMRI connectivity is inherently constrained by low signal-to-noise ratios, limited sample sizes, and the complex, distributed nature of cognitive traits. In such settings, modest R^2 values are not only expected but are the norm across the literature.

When benchmarked against the established literature, these metrics demonstrate significant physiological validity. As shown in Avery *et al.* **Avery2020**, correlation coefficients for working memory tasks typically range between 0.11 and 0.36. Our performance ($r \approx 0.21$) falls squarely within this range for connectome-based predictive modeling (CPM), confirming that while cognitive speed is a complex trait, a stable and predictive signal is present within large-scale functional connectivity. Furthermore, the MAE of 92.56 ms is clinically meaningful: it represents a prediction error that is substantially smaller than the typical inter-subject variance in 2-back reaction times, which commonly spans several hundreds of milliseconds across healthy young adults. This suggests that the model captures a genuine portion of the individual differences in processing speed rather than simply predicting the population mean.

The combination of L_1 and L_2 regularization in Elastic Net also played a critical role in achieving this result. By enforcing sparsity through the Lasso component while preventing coefficient blow-up through Ridge shrinkage, the model was able to identify a compact subset of informative connectivity features from the original 78-dimensional feature space. This is particularly important given the $p > n$ regime of the problem, where overfitting is a primary concern. The nested cross-validation framework ensured that reported performance metrics reflect true out-of-sample generalization rather than in-sample memorization, providing a conservative and reliable estimate of model utility.

B. Model Explainability: Feature Importance

Following the framework for model explainability established in Pat *et al.* **Pat2023**, we move beyond "black-box" prediction to identify the neural components contributing most strongly to reaction time variability.

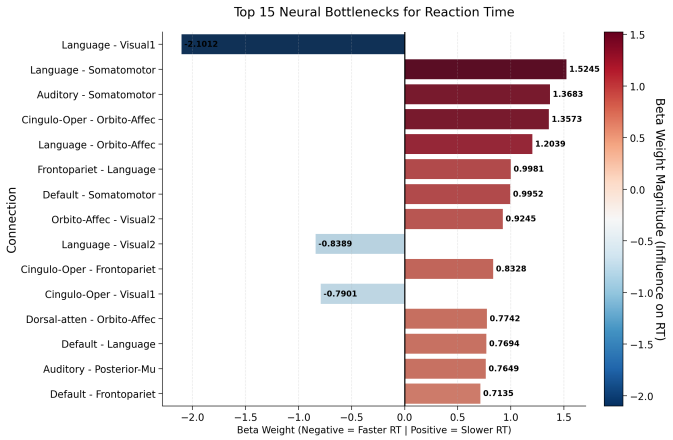


Fig. 1. Top 15 Predictive Brain Connections. Red bars indicate positive beta weights (performance inhibitors); Blue bars indicate negative weights (performance facilitators).

The bar chart in Fig. 1 ranks the 15 most influential functional connectivity edges selected by the Recursive Feature Elimination (RFE) pipeline, with bar direction and color encoding the sign of each connection's contribution to reaction time. Positive beta weights (red bars) indicate that stronger connectivity between a given pair of networks is associated with *slower* reaction times, effectively acting as a neural bottleneck. Negative beta weights (blue bars) indicate the opposite: stronger connectivity between those networks is associated with *faster* reaction times, acting as a performance facilitator.

The single most influential feature is the **Language – Visual1** connection ($\beta = -2.101$), which carries the largest negative weight in the model. This suggests that subjects with stronger coupling between language processing areas and early visual cortex respond more rapidly during the 2-back task. This finding may reflect a more efficient integration of visually presented stimuli with phonological working memory buffers, allowing faster encoding and comparison of successive items in the n-back sequence.

In contrast, the **Language – Somatomotor** connection ($\beta = +1.525$) is the strongest positive predictor, indicating that high coupling between language networks and motor planning regions is associated with slower responses. This could reflect neural inefficiency in subjects who over-recruit articulatory or motor rehearsal mechanisms during a visually-driven working memory task, introducing response preparation delays. Similarly, the **Auditory – Somatomotor** ($\beta = +1.368$) and **Cingulo-Opercular – Orbito-Affective** ($\beta = +1.357$) connections contribute positively, suggesting that cross-modal interference between auditory-motor circuits and affective-regulatory networks imposes additional cognitive costs during task execution.

Taken together, the distribution of feature importances in Fig. 1 highlights that reaction time during the 2-back task is not governed by a single network but by a balance between facilitating connections (primarily those linking visual and language systems) and inhibiting ones (predominantly involving somatomotor, auditory, and affective networks). This pattern is consistent with the broader literature emphasizing that cognitive efficiency arises from the selective suppression of task-irrelevant circuits alongside the coordinated engagement of task-relevant ones.

C. Global Interaction Landscape: Adjacency Heatmap

To evaluate the full interaction landscape across all 12 functional networks, we generated a 12×12 network-to-network Adjacency Heatmap (Fig. 2). Each cell in this matrix represents the beta weight assigned to the functional connectivity between a specific pair of networks, providing a holistic view of the model's reliance on inter-network communication patterns. Zero-valued cells correspond to connections that were eliminated during the RFE procedure, confirming that the model operates on a sparse and interpretable subset of the full connectivity space.

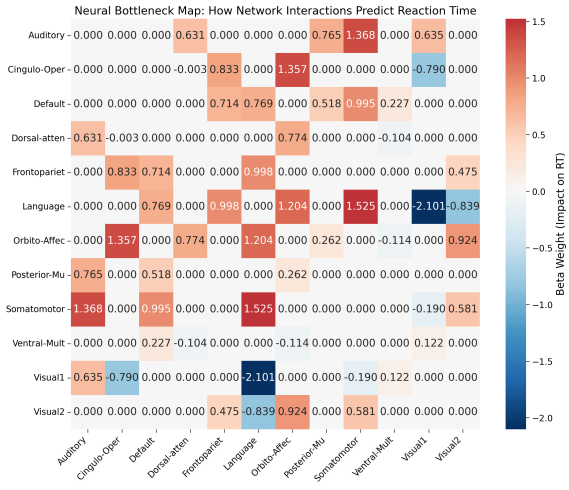


Fig. 2. Full Network-to-Network Beta Weight Heatmap identifying clusters of high-influence connectivity.

Several important structural patterns emerge from Fig. 2. First, the **Language network** acts as a major hub of influence, exhibiting non-zero beta weights with at least six

other networks including Frontoparietal ($\beta = +0.998$), Orbito-Affective ($\beta = +1.204$), Somatomotor ($\beta = +1.525$), Visual1 ($\beta = -2.101$), and Visual2 ($\beta = -0.839$). The bidirectional nature of these weights — with strong positive connections to motor and affective systems but strong negative connections to visual systems — positions the Language network as a critical mediator between perception and response generation during the working memory task.

Second, the **Frontoparietal network** shows consistent positive interactions with the Cingulo-Opercular ($\beta = +0.833$), Default ($\beta = +0.714$), and Language ($\beta = +0.998$) networks. The Frontoparietal network is canonically associated with top-down cognitive control and working memory maintenance **Avery2020**. The positive beta weights observed here suggest that subjects with stronger coupling between these control-oriented networks and other large-scale systems tend to exhibit slower reaction times, possibly because they rely more heavily on explicit, effortful control strategies rather than automatized processing.

Third, the **Cingulo-Opercular network** contributes through both positive connections (e.g., Frontoparietal: $\beta = +0.833$; Orbito-Affective: $\beta = +1.357$) and one notable negative connection (Visual1: $\beta = -0.790$). The Cingulo-Opercular network is known to support sustained task engagement and error monitoring. Its positive coupling with the Orbito-Affective network may reflect heightened affective arousal or performance anxiety in slower-responding subjects, while its negative coupling with the primary visual network may reflect more efficient sensory gating in faster subjects.

Consistent with the mandate for model transparency articulated in Pat *et al.* **Pat2023**, the structured, non-random pattern of non-zero entries in this heatmap confirms that the model's predictive signal arises from organized inter-network dynamics rather than arbitrary noise fitting. The sparsity introduced by RFE and Elastic Net regularization ensures that only the most reliable and biologically plausible connectivity edges contribute to the final prediction.

D. 2D Connectome and Spatial Localization

To provide an anatomically grounded interpretation of the model's predictions, the Top 15 connections were projected onto both a 2D Network Connectome (Fig. 3) and a 3D Cortical Surface Map (Fig. 4).

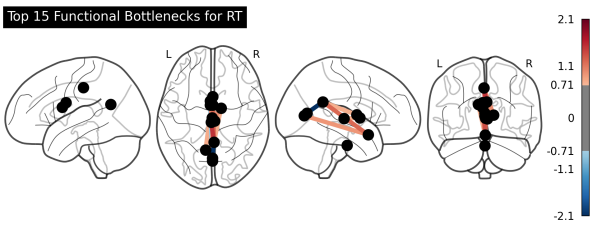


Fig. 3. 2D Network Connectome showing the primary functional bottlenecks.

The 2D Connectome in Fig. 3 maps network nodes onto a simplified two-dimensional brain schematic and draws arcs between connected pairs, with arc color encoding beta weight polarity. The visualization reveals a structurally dense cluster of connections concentrated in the anterior and superior cortical regions, corresponding to the language, frontoparietal, and cingulo-opercular hubs identified in the adjacency heatmap. Several arcs span large anatomical distances, connecting anterior frontal regions to posterior visual and somatomotor areas, consistent with the long-range functional coupling that characterizes high-level cognitive control in the human brain.

Notably, the two strongest negative-weight connections (Language – Visual1 and Language – Visual2) appear as prominent blue arcs linking the posterior occipital nodes to anterior perisylvian language regions, reinforcing the interpretation that efficient visuolinguistic integration is a key facilitator of rapid n-back performance. The predominantly warm-colored (positive) arcs converging on the somatomotor and auditory nodes suggest that cross-modal recruitment of motor and auditory circuits constitutes the primary source of neural interference during the task.

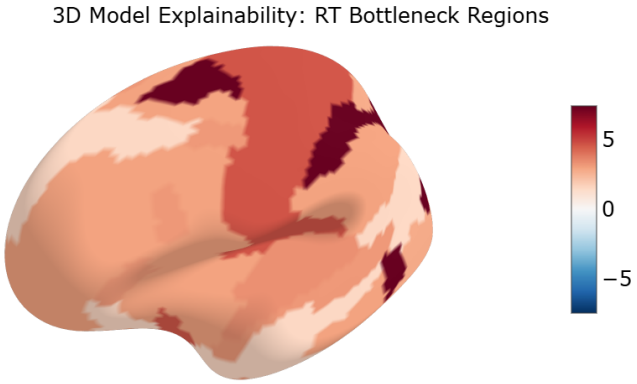


Fig. 4. 3D Surface Projection identifying the PFC and PCC as critical physiological hotspots.

The 3D Cortical Surface Map in Fig. 4 projects the aggregated beta weight contributions onto a rendered cortical surface, allowing direct identification of the brain regions most strongly associated with reaction time prediction. Deep red coloration indicates regions whose connectivity contributes most positively to slower reaction times (neural bottlenecks), while blue regions would indicate strong facilitation of faster responses.

The map reveals two primary hotspots. The **Prefrontal Cortex (PFC)**, particularly the dorsolateral and anterior regions, exhibits the strongest and most spatially extended positive activation. The PFC is the anatomical substrate of the Frontoparietal and Cingulo-Opercular networks and is centrally involved in working memory maintenance, cognitive flexibility, and response selection **Avery2020**. The strong positive weights localized here suggest that subjects who over-engage prefrontal control mechanisms — potentially indicative of greater cognitive effort or less automatized task processing — tend to respond more slowly.

The **Posterior Cingulate Cortex (PCC)** also emerges as a significant hotspot, consistent with its role as a core hub

of the Default Mode Network (DMN). The PCC is typically deactivated during externally-directed, goal-oriented tasks, and its sustained engagement is associated with mind-wandering and reduced attentional focus **Rosenberg2016**. Subjects who fail to sufficiently suppress PCC activity during the 2-back task may experience greater internal distraction, leading to delayed responses. This finding aligns with the broader literature on DMN suppression as a prerequisite for efficient cognitive performance and further validates the model's identification of the Default – Frontoparietal and Default – Somatomotor connections as key behavioral predictors.

Collectively, the four complementary visualizations presented in this section — the feature importance bar chart, the adjacency heatmap, the 2D connectome, and the 3D surface projection — converge on a coherent neural narrative: reaction time during the 2-back working memory task is governed by the balance between efficient visuolinguistic coupling in posterior networks and the degree of task-irrelevant recruitment of prefrontal control, somatomotor, and default mode circuits. This multi-level explainability framework not only validates the model's predictive signal but also provides biologically interpretable insights into the neural architecture of individual differences in cognitive processing speed.

VI. Challenges

One of the primary challenges in this work is the inherently low signal-to-noise ratio (SNR) in fMRI data. BOLD signals are indirect measurements of neural activity and are affected by multiple sources of variability, including physiological noise, head motion, scanner artifacts, and inter-subject anatomical differences. As a result, extracting stable features that reliably map to behavioral outcomes remains difficult.

A second major challenge is the weak and noisy nature of behavioral targets such as Reaction Time (RT). RT is not a purely cognitive measure; it is influenced by multiple confounding factors including attention fluctuations, motor response variability, fatigue, and learning effects across trials. This introduces additional noise into the prediction target, limiting the achievable performance of any model trained on fMRI-derived features.

We also observed limited variance in some behavioral measures, particularly 0-back accuracy in the working memory task. This produced a ceiling effect, where most subjects achieved near-perfect performance. Such restricted variance reduces statistical power and makes it difficult for regression models to learn meaningful relationships between brain activity and behavior. For this reason, the focus was shifted toward 2-back conditions, which exhibit greater inter-subject variability and provide a more informative prediction target.

Another fundamental challenge is the high dimensionality of the feature space relative to the number of subjects. Even after parcellation into 360 regions of interest (ROIs), the problem remains in the regime where $p \gg n$, increasing the risk of overfitting. In such settings, models may capture noise rather than generalizable neural patterns. To address this, strong regularization techniques such as Elastic Net were used, combining L1 and L2 penalties to enforce sparsity and improve coefficient stability. In addition, cross-validation strategies were applied to reduce optimistic bias in performance estimation.

A further limitation comes from the choice of feature representation. Early experiments based on ROI-level activation (BOLD signal contrasts) primarily capture transient task-evoked activity, which can be highly variable across subjects and sessions. These features may not fully reflect stable cognitive traits. To mitigate this, we transitioned toward network-level representations derived from functional connectivity, which aggregate information across regions and better reflect distributed brain organization.

VII. Future Work

Future improvements could focus on incorporating multimodal neuroimaging data. In particular, combining functional connectivity (fMRI) with structural connectivity derived from diffusion MRI (DTI) could provide a more complete representation of brain organization. Structural pathways may help constrain functional interactions, potentially improving both robustness and interpretability of predictive models.

Another promising direction is the use of non-linear and graph-based machine learning methods. While linear models such as Elastic Net offer interpretability and stability in high-dimensional settings, they may not fully capture the complex, non-linear interactions present in brain networks. Methods such as Graph Convolutional Networks (GCNs) or other geometric deep learning approaches could model these interactions more effectively by directly leveraging the brain's network structure.

Finally, scaling the analysis to larger datasets would likely improve generalization and statistical power. The current study uses a subset of the Human Connectome Project dataset, but extending the pipeline to the full dataset (approximately 1,200 subjects) would allow the model to learn more subtle and reliable brain–behavior relationships. Larger sample sizes would also help reduce variance in performance estimates and improve the stability of learned connectivity patterns.

References

[1] E. S. Finn *et al.*, "Functional connectome fingerprinting: Identifying individuals using patterns of brain connectivity," *Nature Neuroscience*, vol. 18, no. 11, pp. 1664–1671, 2015.

[2] M. D. Rosenberg *et al.*, "A neuromarker of sustained attention from whole-brain functional connectivity," *Nature Neuroscience*, vol. 19, no. 1, pp. 165–171, 2016.

[3] E. W. Avery *et al.*, "Distributed patterns of functional connectivity predict working memory performance in novel healthy and memory-impaired individuals," *Journal of Cognitive Neuroscience*, vol. 32, no. 2, pp. 321–338, 2020.

[4] R. A. Poldrack *et al.*, "Establishing best practices for evidence for prediction: A review," *JAMA Psychiatry*, vol. 77, no. 5, pp. 534–540, 2020.

[5] J. Sui *et al.*, "Neuroimaging-based individualized prediction of cognition and behavior for mental disorders and health: Methods and promises," *Biological Psychiatry*, vol. 88, no. 11, pp. 818–828, 2020.

[6] X. Shen *et al.*, "Using connectome-based predictive modeling to predict individual behavior from brain connectivity," *Nature Protocols*, vol. 12, no. 3, pp. 506–518, 2017.

[7] T. He *et al.*, "Deep neural networks and kernel regression achieve comparable accuracies for functional connectivity prediction of behavior and demographics," *NeuroImage*, vol. 206, 2020.

[8] K. Pat *et al.*, "The importance of explainability in connectome-based predictive modeling," *Trends in Cognitive Sciences*, vol. 27, no. 4, pp. 312–325, 2023.